Networking – IPv6-Address

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Introduction of IPV6

- Internet Protocol Version 6(IPv6) is the latest revision of the Internet Protocol, the communication protocol that provides an identification and location system for computers on networks and routes traffic across the internet.

- IPv6 was developed by IETF to deal with the long-anticipated problem of IPv4 address exhaustion.

- IPv6 was intended to replace IPv4, which still carries the vast majority of Internet traffic.
- In contrast to IPv4, which defined an IP address as a 32-bit value, IPv6 addresses have a size of 128 bits. Therefore, IPv6 has a vastly enlarged address space compared to IPv4.

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Features of IPV6

- New Header Format
- Large Address Space
- Efficient and Hierarchical addressing and routing infrastructure
- Stateless and stateful address configuration
- Built-in Security
- Better support for Quality of Service
- New support for neighboring node interaction
- Extensibility

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IPV6 Addressing

The number of possible address in IPv6 is 2^128(3.4*10^38) addresses.

340,282,366,920,938,463,463,374,607,431,768,211,456 to be exact.

Such a large amount of available IP addresses ensure that we will never again run out of IP addresses.

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IPV6 Addressing

- The IPv6 address size is 128-bit which is represented in hexadecimals.
- The 128-bit address is divided into 16-bits, and each 16-bit block is converted into 4-digit hexadecimal number and separated by colons. This type o f representation is called colon hexadecimal.
- The format o f IPv6 address is

xxxx: xxxx: xxxx: xxxx: xxxx: xxxx: xxxx: xxxx

where each **X** is a hexadecimal representing **4 bits** or a nibble.

- IPv6 address range from
- 0000:0000:0000:0000:0000:0000:0000:0000 to
  ffff: ffff: ffff: ffff: ffff: ffff: ffff: ffff.

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IPV6 Addressing

IPV6 is an 128 Bit Address

Its an HEXA Decimal Values.

2600:ABCD:EF01:0000:0000:ABCD:EF01:2345 → IPv6 Example.

Each digit is an HEXA Decimal number. And has “4” bits.

2600 representing in Binary → 0010 0110 0000 0000

Points.

- If the Address is starting with “**2**” then its an public IPV6 address.
- If the Address is starting with “**FE80**” then its an LINK-Local IPV6 address

HexaDecimal	-- Binary
0	-- 0000
1	-- 0001
2	-- 0010
3	-- 0011
.	
.	
9	-- 1001
A	-- 1010
B	-- 1011
C	-- 1100
.	
F	-- 1111

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IPV6 Addressing

IPv6 address can be simplified by following two methods :

- **Omit Leading Zeros**

For Example, 4FDE:0000:0000:0002:0022:F376:FF3B:AC99 may be written as 4FDE:0:0:2:22:F376:FF3B:AC99.

Eg2 -- 4FDE:0000:0000:0002:0022:F376:0000:AC99

4FDE:0:0:2:22:F376:0:AC99.

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IPV6 Addressing

IPv6 address can be simplified by following two methods :

- **Double Colon**

For Example, 4FDE:0000:0000:0002:0022:F376:FF3B:AC99 can be further simplified as **4FDE::2:22:F376:FF3B:AC99**.

Eg2:-- 4FDE:0000:0000:0000:0022:F376:FF3B:AC99

Can be written as -- **4FDE::22:F376:FF3B:AC99**.

Eg3:-- 4FDE:0000:0000:0022:F376:0000:0000:AC99

Can be written as -- **4FDE::22:F376:FF3B:0:0:AC99**.

Or --- **4FDE:0:0:22:F376:FF3B::AC99**.

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Reserved Ranges of IPV6

FF00::/8 – Multicast IPV6 range

2000::/8 – Public IPV6 range

FE80::/12 – Link Local IPV6 Range

FD00::/10 – Private IPV6 range.

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IPV6 Addressing

- **Subnetting** concepts remain the same as IPV4
- There is NO concept of **PATTING** as have lots of IP addresses that can be spared in IPV6.
- BUT **NATTING** concept exists for HIDDING the server.

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IPV6 Address -- Subnetting

- **Given Main CIDR network**
- **2600:1f18:60e1:d300**::/**96**
- Also be represented as **2600:1f18:60e1:d300:0:0**::/**96**
- Means **96** bit are **network bit's** and remaining **32** are Host bits
- **H = 128 -96 = 32 bits**

- **Requirement for subnetting**
- **We need 1000 ips per network and we need 50 networks**

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IPV6 Address -- Subnetting

- **Solution:**
- **1000 ~ 2^10**
- But since we have lots of IPV6, we would round off the binary's to 4 bits at a time.
- Either we can say **– New H Bits = 12 bits or even 16 bits for our easy.**
- Which means → **New N Bits = 128 - 16 = 112**
- **The First Subnetwork → 2600:1f18:60e1:d300:0:0:0::/112**
- **Second Subnetwork → 2600:1f18:60e1:d300:0:0:1::/112**
- **Third Subnetwork → 2600:1f18:60e1:d300:0:0:2::/112**

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This is the 7th 16bit set in the previous example, which would keep increasing by binary "1" to get the next subnetwork.

0000 0000 0000 0000 → 0000 → 0

0000 0000 0000 0001 → 0001 → 1

0000 0000 0000 0010 → 0002 → 2

0000 0000 0000 0011 → 0003 → 3

……

0000 0000 1111 0100 → 00F4 → F4

……

0000 1111 0000 0101 → 0F05 → F05

….

1111 1111 1111 1111 → FFFF

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IPV4 V/S IPV6

IPv4	IPv6
32- bit length.	128-bit length
IPSec support is only optional.	Inbuilt IPSec support
Fragmentation is done by sender and forwarding routers	Fragmentation is only done by sender
No packet flow identification	Packet flow identification is available using flow label field
Checksum field is available in header	No checksum field in header
Option fields are available in header.	No option fields, but extension headers are available.
Broadcast messages are possible	Broadcast messages are not possible.

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IPv6 on AWS

Example -- 2600:1f18:60e1:d3 00::/56 – VPC CIDR Ipv6 Public
/56 is Network and Host Bits are –> 128 -56 = 72 bits , means 2^72 number of HOSTs can be created in this network.

For every Subnet in the VPC

2600:1f18:60e1:d300000000::/64 – First Subnet for IPV6

2600:1f18:60e1:d300000001::/64
2600:1f18:60e1:d300000010::/64

2600:1f18:60e1:d300000011::/64

2600:1f18:60e1:d31111 1111::/64 -- **2600:1f18:60e1:d3 FF**::/64 – Last IPv6 Subnetwork that can be used.

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More Reference Links on IPV6

https://www.mediawiki.org/wiki/Help:Range_blocks/IPv6

https://www.iana.org/assignments/ipv6-address-space/ipv6-address-space.xhtml

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